Advanced Trigonometry

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Trigonometric identities: Pythagorean and reciprocal**

Given $\cos\theta = \frac{4}{5}$ and θ in the fourth quadrant, find $\sin\theta$ and $\tan\theta$.

$\sin^2\theta = 1 - \frac{16}{25} = \frac{9}{25}$. In the fourth quadrant, sine is negative: $\sin\theta = -\frac{3}{5}$. $\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{-3/5}{4/5} = -\frac{3}{4}$.

2 **[Foundational]** • **Target: Graphs and transformations of trigonometric functions**

State the amplitude, period, and phase shift of $y = 2\sin\left(3x - \frac{\pi}{2}\right)$.

Amplitude: 2. Period: $\frac{2\pi}{3}$. Phase shift: solve $3x - \frac{\pi}{2} = 0$, giving $x = \frac{\pi}{6}$ (shift right by $\frac{\pi}{6}$).

3 **[Foundational]** • **Target: Applications: Law of Sines and Law of Cosines together**

Two ships leave a port at the same time, one traveling at 20° and the other at 80° (measured from north), each at 15 mph. Find the distance between the ships after 2 hours, to the nearest mile, using the Law of Cosines.

Each ship travels $15(2) = 30$ miles. The angle between their paths: $80^\circ - 20^\circ = 60^\circ$. Using the Law of Cosines: $d^2 = 30^2 + 30^2 - 2(30)(30)\cos 60^\circ = 900 + 900 - 1800(0.5) = 1800 - 900 = 900$. $d = \sqrt{900} = 30$ miles.

4 **[Foundational]** • **Target: Polar coordinates**

Convert the polar point $\left(4, \frac{\pi}{3}\right)$ to rectangular coordinates.

$x = r\cos\theta = 4\cos\left(\frac{\pi}{3}\right) = 4\left(\frac{1}{2}\right) = 2$. $y = r\sin\theta = 4\sin\left(\frac{\pi}{3}\right) = 4\left(\frac{\sqrt{3}}{2}\right) = 2\sqrt{3}$. Rectangular coordinates: $(2, 2\sqrt{3})$.

5 **[Foundational]** • **Target: Area of a triangle using trigonometry**

Find the area of a triangle with sides 8 and 12, and included angle 50°, using $\text{Area} = \frac{1}{2}ab\sin C$.

$\text{Area} = \frac{1}{2}(8)(12)\sin(50^\circ) \approx \frac{1}{2}(96)(0.766) \approx 36.8$.

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6 [Foundational] • Target: A multi-part navigation and triangulation problem

This question has three parts. A hiker walks 5 miles due east, then turns and walks 7 miles at a bearing of 40° north of east. (a) Find the angle between the two legs of the hike (measured as the interior angle at the turning point). (b) Use the Law of Cosines to find the straight-line distance from the start to the end point, to the nearest tenth of a mile. (c) Explain why this distance is less than the total distance walked (12 miles).

(a) The second leg deviates 40° from due east, so the interior angle at the turn (between reversing the first leg's direction and the second leg) is $180^\circ - 40^\circ = 140^\circ$. (b) $d^2 = 5^2 + 7^2 - 2(5)(7)\cos(140^\circ) = 25 + 49 - 70(-0.766) \approx 74 + 53.6 \approx 127.6$. $d \approx \sqrt{127.6} \approx 11.3$ miles. (c) The straight-line distance is always less than or equal to the sum of the individual leg lengths (a consequence of the triangle inequality) — walking in two different directions doesn't add distance as efficiently as walking directly toward the destination, since the path bends rather than going straight.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Given $\cos\theta = \frac{4}{5}$ and θ in the fourth quadrant, find $\sin\theta$ and $\tan\theta$.

$\sin^2\theta = 1 - \frac{16}{25} = \frac{9}{25}$. In the fourth quadrant, sine is negative: $\sin\theta = -\frac{3}{5}$. $\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{-3/5}{4/5} = -\frac{3}{4}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

State the amplitude, period, and phase shift of $y = 2\sin\left(3x - \frac{\pi}{2}\right)$.

Amplitude: 2. Period: $\frac{2\pi}{3}$. Phase shift: solve $3x - \frac{\pi}{2} = 0$, giving $x = \frac{\pi}{6}$ (shift right by $\frac{\pi}{6}$).

9 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the exact value of $\arccos\left(-\frac{\sqrt{2}}{2}\right)$.

$\arccos\left(-\frac{\sqrt{2}}{2}\right) = \frac{3\pi}{4}$ (or 135°), since $\cos\left(\frac{3\pi}{4}\right) = -\frac{\sqrt{2}}{2}$ and $\frac{3\pi}{4}$ is in the range of arccosine.

10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Convert the polar point $\left(4, \frac{\pi}{3}\right)$ to rectangular coordinates.

$x = r\cos\theta = 4\cos\left(\frac{\pi}{3}\right) = 4\left(\frac{1}{2}\right) = 2$. $y = r\sin\theta = 4\sin\left(\frac{\pi}{3}\right) = 4\left(\frac{\sqrt{3}}{2}\right) = 2\sqrt{3}$. Rectangular coordinates: $(2, 2\sqrt{3})$.

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11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

This question has two parts. (a) Use the identity $\tan\theta = \frac{\sin\theta}{\cos\theta}$ together with the double angle formulas to derive a formula for $\tan(2\theta)$ in terms of $\tan\theta$. (b) Use your formula to find $\tan(2\theta)$ given $\tan\theta = 3$.

(a) $\tan(2\theta) = \frac{\sin(2\theta)}{\cos(2\theta)} = \frac{2\sin\theta\cos\theta}{\cos^2\theta - \sin^2\theta}$. Dividing numerator and denominator by $\cos^2\theta$: $\tan(2\theta) = \frac{2\tan\theta}{1-\tan^2\theta}$.
(b) $\tan(2\theta) = \frac{2(3)}{1-9} = \frac{6}{-8} = -\frac{3}{4}$.

12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the area of a triangle with sides 8 and 12, and included angle 50°, using $\text{Area} = \frac{1}{2}ab\sin C$.

$\text{Area} = \frac{1}{2}(8)(12)\sin(50^\circ) \approx \frac{1}{2}(96)(0.766) \approx 36.8$.

13 [Intermediate] • Target: Sum and difference identities

Use the sum identity to find the exact value of $\sin(75^\circ)$, using $\sin(75^\circ) = \sin(45^\circ + 30^\circ)$.

$\sin(45^\circ + 30^\circ) = \sin 45^\circ\cos 30^\circ + \cos 45^\circ\sin 30^\circ = \frac{\sqrt{2}}{2}\cdot\frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2}\cdot\frac{1}{2} = \frac{\sqrt{6}+\sqrt{2}}{4}$.

14 [Intermediate] • Target: Double angle identities

Given $\sin\theta = \frac{3}{5}$ and θ in the first quadrant, find $\sin(2\theta)$ and $\cos(2\theta)$.

$\cos\theta = \frac{4}{5}$ (first quadrant, positive). $\sin(2\theta) = 2\sin\theta\cos\theta = 2\left(\frac{3}{5}\right)\left(\frac{4}{5}\right) = \frac{24}{25}$.
$\cos(2\theta) = 1 - 2\sin^2\theta = 1 - 2\left(\frac{9}{25}\right) = 1 - \frac{18}{25} = \frac{7}{25}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Inverse trigonometric functions • Cross-Lesson: 5.01 Pre-Calculus Concepts

Find the exact value of $\arcsin\left(\frac{1}{2}\right)$.

$\arcsin\left(\frac{1}{2}\right) = \frac{\pi}{6}$ (or 30°), since $\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$ and $\frac{\pi}{6}$ is in the range of arcsine.

16 [Challenging] • Target: Inverse trigonometric functions • Cross-Lesson: 6.01 Advanced Coordinate Geometry

Find the exact value of $\arccos\left(-\frac{\sqrt{2}}{2}\right)$.

$\arccos\left(-\frac{\sqrt{2}}{2}\right) = \frac{3\pi}{4}$ (or 135°), since $\cos\left(\frac{3\pi}{4}\right) = -\frac{\sqrt{2}}{2}$ and $\frac{3\pi}{4}$ is in the range of arccosine.

17 [Challenging] • Target: Trigonometric equations with multiple solutions • Cross-Lesson: 7.01 Advanced Plane Geometry

Solve $2\cos^2\theta - 1 = 0$ for $0^\circ \le \theta < 360^\circ$.

$\cos^2\theta = \frac{1}{2}$, so $\cos\theta = \pm\frac{\sqrt{2}}{2}$. Solutions: $\theta = 45^\circ, 135^\circ, 225^\circ, 315^\circ$.

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18 [Challenging] • Target: A multi-part trigonometric identity verification problem • Cross-Lesson: 8.01 Statistics, Counting & Probability

This question has two parts. (a) Use the identity $\tan\theta = \frac{\sin\theta}{\cos\theta}$ together with the double angle formulas to derive a formula for $\tan(2\theta)$ in terms of $\tan\theta$. (b) Use your formula to find $\tan(2\theta)$ given $\tan\theta = 3$.

(a) $\tan(2\theta) = \frac{\sin(2\theta)}{\cos(2\theta)} = \frac{2\sin\theta\cos\theta}{\cos^2\theta - \sin^2\theta}$. Dividing numerator and denominator by $\cos^2\theta$: $\tan(2\theta) = \frac{2\tan\theta}{1-\tan^2\theta}$. (b) $\tan(2\theta) = \frac{2(3)}{1-9} = \frac{6}{-8} = -\frac{3}{4}$.

19 [Challenging] • Target: Half-angle identities • Cross-Lesson: 1.01 Advanced Number Theory

Use the half-angle identity to find $\cos(15^\circ)$, given $\cos(30^\circ) = \frac{\sqrt{3}}{2}$.

Half-angle identity: $\cos\left(\frac{\theta}{2}\right) = \sqrt{\frac{1+\cos\theta}{2}}$. With $\theta = 30^\circ$: $\cos(15^\circ) = \sqrt{\frac{1+\frac{\sqrt{3}}{2}}{2}} = \sqrt{\frac{2+\sqrt{3}}{4}} = \frac{\sqrt{2+\sqrt{3}}}{2}$.

20 [Challenging] • Target: Solving trigonometric equations that factor • Cross-Lesson: 2.01 Advanced Algebra & Systems

Solve $2\sin^2\theta - \sin\theta - 1 = 0$ for $0^\circ \le \theta < 360^\circ$.

This factors as a quadratic in $\sin\theta$: $(2\sin\theta + 1)(\sin\theta - 1) = 0$, so $\sin\theta = -\frac{1}{2}$ or $\sin\theta = 1$. For $\sin\theta = 1$: $\theta = 90^\circ$. For $\sin\theta = -\frac{1}{2}$: $\theta = 210^\circ$ or $\theta = 330^\circ$. All solutions: $90^\circ, 210^\circ, 330^\circ$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Trigonometric identities: Pythagorean and reciprocal

Given $\sin\theta = \frac{15}{17}$ and θ in the second quadrant, find $\cos\theta$.

a. $-\frac{15}{17}$

b. $\frac{15}{17}$

c. $-\frac{8}{17}$ ✓

d. $\frac{8}{17}$

$\sin^2\theta + \cos^2\theta = 1$, so $\cos\theta^2 = 1 - \sin\theta^2 = 1 - 225/289 = 64/289$. In the second quadrant, cosine is negative: $\cos\theta = -8/17$.

22 [Foundational] • Target: Sum and difference identities

Given $\cos\theta = -\frac{4}{5}$ and θ in the second quadrant, find $\sin\theta$.

a. $\frac{4}{5}$

b. $\frac{3}{5}$ ✓

c. $-\frac{4}{5}$

d. $-\frac{3}{5}$

$\sin^2\theta + \cos^2\theta = 1$, so $\sin\theta^2 = 1 - \cos\theta^2 = 1 - 16/25 = 9/25$. In the second quadrant, sine is positive: $\sin\theta = 3/5$.

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23 [Foundational] • Target: Double angle identities

Given $\cos\theta = -\frac{7}{25}$ and θ in the second quadrant, find $\sin\theta$.

a. $\frac{24}{25}$ ✓

b. $-\frac{24}{25}$

c. $\frac{7}{25}$

d. $-\frac{7}{25}$

$\sin^2\theta + \cos^2\theta = 1$, so $\sin\theta^2 = 1 - \cos\theta^2 = 1 - 49/625 = 576/625$. In the second quadrant, sine is positive: $\sin\theta = 24/25$.

24 [Foundational] • Target: Graphs and transformations of trigonometric functions

Given $\cos\theta = \frac{12}{37}$ and θ in the first quadrant, find $\sin\theta$.

a. $\frac{12}{37}$

b. $\frac{35}{37}$ ✓

c. $-\frac{35}{37}$

d. $-\frac{12}{37}$

$\sin^2\theta + \cos^2\theta = 1$, so $\sin\theta^2 = 1 - \cos\theta^2 = 1 - 144/1369 = 1225/1369$. In the first quadrant, sine is positive: $\sin\theta = 35/37$.

25 [Foundational] • Target: Inverse trigonometric functions

Given $\cos\theta = -\frac{8}{17}$ and θ in the second quadrant, find $\sin\theta$.

a. $\frac{8}{17}$

b. $-\frac{8}{17}$

c. $-\frac{15}{17}$

d. $\frac{15}{17}$ ✓

$\sin^2\theta + \cos^2\theta = 1$, so $\sin\theta^2 = 1 - \cos\theta^2 = 1 - 64/289 = 225/289$. In the second quadrant, sine is positive: $\sin\theta = 15/17$.

26 [Foundational] • Target: Trigonometric equations with multiple solutions

Given $\cos\theta = -\frac{35}{37}$ and θ in the third quadrant, find $\sin\theta$.

a. $\frac{12}{37}$

b. $-\frac{35}{37}$

c. $-\frac{12}{37}$ ✓

d. $\frac{35}{37}$

$\sin^2\theta + \cos^2\theta = 1$, so $\sin\theta^2 = 1 - \cos\theta^2 = 1 - 1225/1369 = 144/1369$. In the third quadrant, sine is negative: $\sin\theta = -12/37$.

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27 [Foundational] • Target: Applications: Law of Sines and Law of Cosines together

Given $\cos\theta = \frac{12}{13}$ and θ in the first quadrant, find $\sin\theta$.

a. $-\frac{5}{13}$

b. $-\frac{12}{13}$

c. $\frac{5}{13}$ ✓

d. $\frac{12}{13}$

$\sin^2\theta + \cos^2\theta = 1$, so $\sin\theta^2 = 1 - \cos\theta^2 = 1 - 144/169 = 25/169$. In the first quadrant, sine is positive: $\sin\theta = 5/13$.

28 [Foundational] • Target: Polar coordinates

Solve $2\cos\theta = \sqrt{2}$ for $0^\circ \le \theta < 360^\circ$.

a. $135^\circ, 225^\circ$

b. $225^\circ, 315^\circ$

c. $45^\circ, 315^\circ$ ✓

d. $45^\circ, 135^\circ$

$\cos\theta = \frac{\sqrt{2}}{2}$. The cosine of θ equals this value at $\theta = 45^\circ, 315^\circ$ within $0^\circ \le \theta < 360^\circ$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A multi-part trigonometric identity verification problem

Solve $\sin\theta = 1$ for $0^\circ \le \theta < 360^\circ$.

a. $225^\circ, 315^\circ$

b. $45^\circ, 135^\circ$

c. 90° ✓

d. 0°

$\sin\theta = 1$. The sine of θ equals this value at $\theta = 90^\circ$ within $0^\circ \le \theta < 360^\circ$.

30 [Intermediate] • Target: Half-angle identities

Solve $2\sin\theta = -\sqrt{3}$ for $0^\circ \le \theta < 360^\circ$.

a. $30^\circ, 330^\circ$

b. $240^\circ, 300^\circ$ ✓

c. $120^\circ, 240^\circ$

d. $60^\circ, 300^\circ$

$\sin\theta = -\frac{\sqrt{3}}{2}$. The sine of θ equals this value at $\theta = 240^\circ, 300^\circ$ within $0^\circ \le \theta < 360^\circ$.

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31 [Intermediate] • Target: Solving trigonometric equations that factor

Solve $2\sin\theta = -\sqrt{2}$ for $0^\circ \le \theta < 360^\circ$.

a. $120^\circ, 240^\circ$

b. $45^\circ, 135^\circ$

c. 0°

d. $225^\circ, 315^\circ$ ✓

$\sin\theta = -\frac{\sqrt{2}}{2}$. The sine of θ equals this value at $\theta = 225^\circ, 315^\circ$ within $0^\circ \le \theta < 360^\circ$.

32 [Intermediate] • Target: Area of a triangle using trigonometry

Solve $2\sin\theta = \sqrt{3}$ for $0^\circ \le \theta < 360^\circ$.

a. $60^\circ, 120^\circ$ ✓

b. 180°

c. 90°

d. $30^\circ, 330^\circ$

$\sin\theta = \frac{\sqrt{3}}{2}$. The sine of θ equals this value at $\theta = 60^\circ, 120^\circ$ within $0^\circ \le \theta < 360^\circ$.

33 [Intermediate] • Target: Simplifying trigonometric expressions using identities

Solve $2\cos\theta = -1$ for $0^\circ \le \theta < 360^\circ$.

a. $135^\circ, 225^\circ$

b. 180°

c. $120^\circ, 240^\circ$ ✓

d. $150^\circ, 210^\circ$

$\cos\theta = -\frac{1}{2}$. The cosine of θ equals this value at $\theta = 120^\circ, 240^\circ$ within $0^\circ \le \theta < 360^\circ$.

34 [Intermediate] • Target: A multi-part navigation and triangulation problem

Given $\sin\theta = \frac{20}{29}$, find $\cos(2\theta)$.

a. $\frac{840}{841}$

b. $-\frac{41}{841}$

c. $\frac{41}{841}$ ✓

d. $-\frac{840}{841}$

$\cos(2\theta) = 1 - 2\sin^2\theta = 1 - 2\left(\frac{20}{29}\right)^2 = 1 - \frac{800}{841} = \frac{41}{841}$.

35 [Intermediate] • Target: Verifying a trigonometric identity

Given $\sin\theta = \frac{5}{13}$, find $\cos(2\theta)$.

a. $-\frac{120}{169}$

b. $\frac{119}{169}$ ✓

c. $\frac{120}{169}$

d. $-\frac{119}{169}$

$\cos(2\theta) = 1 - 2\sin^2\theta = 1 - 2\left(\frac{5}{13}\right)^2 = 1 - \frac{50}{169} = \frac{119}{169}$.

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36 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Two sides of a triangle measure 16 and 16, with an included angle of 60°. Find the length of the third side, using the Law of Cosines.

a. 23

b. 16 ✓

c. 15

d. 17

Using the Law of Cosines with $\cos(60^\circ) = \frac{1}{2}$:
$c^2 = 16^2 + 16^2 - 2(16)(16)\left(\frac{1}{2}\right) = 256 + 256 - 256 = 256$, so $c = \sqrt{256} = 16$.

37 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Two sides of a triangle measure 3 and 8, with an included angle of 60°. Find the length of the third side, using the Law of Cosines.

a. 8

b. 7 ✓

c. 6

d. 9

Using the Law of Cosines with $\cos(60^\circ) = \frac{1}{2}$: $c^2 = 3^2 + 8^2 - 2(3)(8)\left(\frac{1}{2}\right) = 9 + 64 - 24 = 49$, so $c = \sqrt{49} = 7$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Statistics, Counting & Probability

Find the area of a triangle with sides 12 and 12, and included angle 150°, using $A = \frac{1}{2}ab\sin C$.

a. 72

b. $-36\sqrt{3}$

c. 37

d. 36 ✓

$A = \frac{1}{2}(12)(12)\sin(150^\circ) = \frac{144}{2} \cdot \frac{1}{2} = 36$.

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39 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Advanced Number Theory

Find the area of a triangle with sides 14 and 9, and included angle 150°, using $A = \frac{1}{2}ab\sin C$.

a. $-\frac{63}{2}\sqrt{3}$

b. 63

c. $\frac{63}{2}$ ✓

d. $\frac{65}{2}$

$A = \frac{1}{2}(14)(9)\sin(150^\circ) = \frac{126}{2} \cdot \frac{1}{2} = \frac{63}{2}$.

40 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.01 Advanced Algebra & Systems

Given $\cos\theta = -\frac{4}{5}$ and θ in the second quadrant, find $\sin\theta$.

a. $\frac{4}{5}$

b. $\frac{3}{5}$ ✓

c. $-\frac{4}{5}$

d. $-\frac{3}{5}$

$\sin^2\theta + \cos^2\theta = 1$, so $\sin\theta^2 = 1 - \cos\theta^2 = 1 - 16/25 = 9/25$. In the second quadrant, sine is positive: $\sin\theta = 3/5$.

41 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Advanced Functions

Given $\cos\theta = -\frac{12}{13}$ and θ in the third quadrant, find $\sin\theta$.

a. $-\frac{12}{13}$

b. $\frac{12}{13}$

c. $-\frac{5}{13}$ ✓

d. $\frac{5}{13}$

$\sin^2\theta + \cos^2\theta = 1$, so $\sin\theta^2 = 1 - \cos\theta^2 = 1 - 144/169 = 25/169$. In the third quadrant, sine is negative: $\sin\theta = -5/13$.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 5.01 Pre-Calculus Concepts

Given $\cos\theta = \frac{24}{25}$ and θ in the first quadrant, find $\sin\theta$.

a. $\frac{7}{25}$ ✓

b. $-\frac{24}{25}$

c. $-\frac{7}{25}$

d. $\frac{24}{25}$

$\sin^2\theta + \cos^2\theta = 1$, so $\sin\theta^2 = 1 - \cos\theta^2 = 1 - 576/625 = 49/625$. In the first quadrant, sine is positive: $\sin\theta = 7/25$.

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.01 Advanced Coordinate Geometry

Given $\sin\theta = \frac{5}{13}$, find $\cos(2\theta)$.

a. $-\frac{120}{169}$

b. $\frac{119}{169}$ ✓

c. $\frac{120}{169}$

d. $-\frac{119}{169}$

$\cos(2\theta) = 1 - 2\sin^2\theta = 1 - 2\left(\frac{5}{13}\right)^2 = 1 - \frac{50}{169} = \frac{119}{169}$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Advanced Plane Geometry

If $\sin\theta = \frac{3}{5}$ and θ is in Quadrant II, what is the exact value of $\sin(2\theta)$?

a. $-\frac{24}{25}$ ✓

b. $\frac{24}{25}$

c. $-\frac{7}{25}$

d. $\frac{7}{25}$

In Quadrant II, $\cos\theta = -\sqrt{1 - (3/5)^2} = -\frac{4}{5}$. Then $\sin(2\theta) = 2\sin\theta\cos\theta = 2(\frac{3}{5})(-\frac{4}{5}) = -\frac{24}{25}$.

45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 8.01 Statistics, Counting & Probability

Solve for $x \in [0, \pi]$: $\sqrt{3}\tan(2x) = 1$.

a. $\frac{\pi}{12}, \frac{7\pi}{12}$ ✓

b. $\frac{\pi}{6}, \frac{5\pi}{6}$

c. $\frac{\pi}{3}$

d. $\frac{\pi}{4}, \frac{3\pi}{4}$

$\tan(2x) = \frac{1}{\sqrt{3}}$. For $2x \in [0, 2\pi]$, $2x = \frac{\pi}{6}$ or $2x = \frac{7\pi}{6} \Rightarrow x = \frac{\pi}{12}$ or $x = \frac{7\pi}{12}$.

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